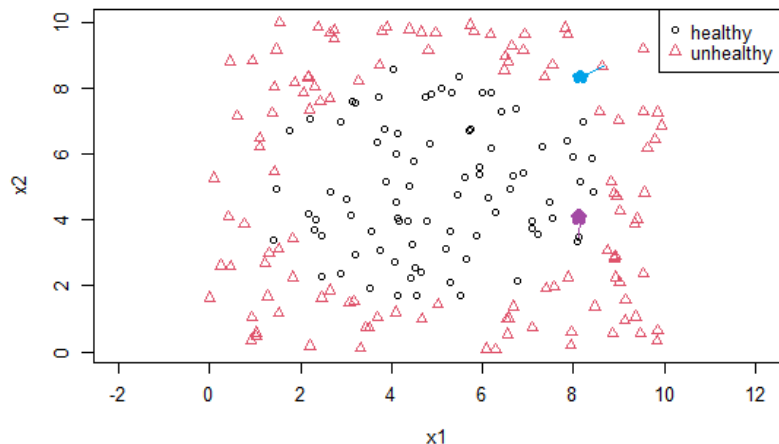


k-nearest neighbors

Albert Y. Kim

k-nearest neighbors

A simple rule is to classify each new data-point using its *nearest neighbor*, or the observation closest to it's x_2 and x_1 coordinates:



Quantifying error

- ▶ This example uses a binary categorical outcome y , so *classification accuracy* is a good measure of error. Later we'll consider other **metrics** for binary outcomes y
- ▶ For a numeric outcome, it's most natural to measure error by summarizing the distances between predicted and observed outcomes $\hat{y}_i$ and y_i (where $\hat{y}_i$ our model's prediction for the i^{th} data-point):
 - ▶ **Mean Squared Error:** $MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
 - ▶ **Root Mean Squared Error:** $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
 - ▶ **Mean Absolute Error:** $MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$

Minkowski distance

To implement kNN, we need to define how to determine the nearest neighbor i.e. what is the distance between points $\mathbf{x}_a$ and $\mathbf{x}_b$.

We'll use Minkowski's distance, $d_{a,b}$, to measure the distance for a hyperparameter p you choose

$$d(\mathbf{x}_1, \mathbf{x}_2) = \left(\sum_{j=1}^{p^*} |x_{1,j} - x_{2,j}|^p \right)^{1/p}$$

Unfortunately this is the same letter p we've been using to describe the number of predictors; in the formula above we represent the number of predictors with p^*

Popular distance measures

Two of the most popular choices are

► $p = 2$:

$$d_{\text{euclidean}} = \sqrt{\sum_{j=1}^{p^*} (x_{a,j} - x_{b,j})^2}$$

► $p = 1$:

$$d_{\text{manhattan}} = \sum_{j=1}^{p^*} |x_{a,j} - x_{b,j}|$$

Other hyperparameters (weighting)

There are two schemes by which neighbors can contribute to a prediction:

1. *Uniform weighting* - each neighbor contributes equally
2. *Distance weighting* - neighbors are weighted by the inverse of their distance to the new data-point, allowing closer neighbors to contribute more strongly

For

- ▶ Regression tasks i.e. y numerical the prediction is either a simple or weighted average
- ▶ Classification tasks i.e. y categorical the predicted probabilities are given by either simple or weighted voting